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ORIGINAL ARTICLE

Prevalence of urinary incontinence in high-impact sports athletes and their association with knowledge, attitude and practice about this dysfunction

AMANDA MARIA BRITO CARDOSO¹, CLÁUDIA REGINA OLIVEIRA DE PAIVA LIMA²,
& CAROLINE WANDERLEY SOUTO FERREIRA ¹

¹Physiotherapy Department, Federal University of Pernambuco, Recife, Brazil & ²Statistics Department, Federal University of Pernambuco, Recife, Brazil

ABSTRACT

Objective: To evaluate the prevalence of urinary incontinence (UI) in female athletes practising high-impact sports and its association with knowledge, attitude and practice (KAP).

Design: Observational study.

Methods: A semi-structured evaluation form, KAP survey and the International Consultation on Incontinence Questionnaire – Short Form were used. Univariate and multivariate analyses were performed to verify the association of UI with KAP, sociodemographic, gynaecological and sports-related variables, with significant variables remaining in the model ($p < .05$), through a stepwise strategy.

Results: The prevalence of UI in 118 athletes was 82 (70%), of which 19 (23%) had pure stress UI, 19 (23%) had pure urgency UI and 44 (54%) had mixed UI; with 50 (61%) athletes losing urine during training and 37 (45%) during competitions. Regarding the KAP survey, 37 athletes (31%) demonstrated adequate knowledge, 63 (53%) adequate attitude and zero adequate practice. In the multivariate analysis, a sports practice time of over 8 years and dyspareunia presented significant values, with odds ratios (OR) of 2.66 and 2.99, respectively. Knowledge was significantly associated with UI (OR 0.43). UI had a slight impact on QOL.

Conclusions: There is a high prevalence of UI yet low levels of adequate knowledge and practice regarding the occurrence of UI in sports, with sports practice time over 8 years and dyspareunia factors predisposing female athletes to UI by 2.7 and 3 times, respectively. Athletes with adequate knowledge were 57% less likely to develop UI.

Keywords: Urinary incontinence, health knowledge, attitudes, practice, exercise, athletes, prevalence

Highlights

- Mixed urinary incontinence is more prevalent than pure stress urinary incontinence and urgency urinary incontinence in female high-impact sports athletes.
- Despite the high prevalence of urinary incontinence, few athletes sought medical attention, none sought physiotherapeutic care and none reported urinary loss to the trainer.
- The practice of strengthening pelvic floor muscle is poorly performed by athletes, due to the lack of knowledge of the existence of these exercises.
- It is recommended for future training practice, to introduce preventive physiotherapy with the aim of evaluating the pelvic floor muscle; as well as, teaching exercises to strengthen fast and slow twitch muscle fibers of the pelvic floor muscle.

Introduction

Regular practice of physical activity is recommended because it provides numerous health benefits. However, physical exercise, especially high-impact sports, has been identified as a risk factor for

urinary incontinence (UI) (Parmigiano et al., 2014; Vitton et al., 2011).

The definition of UI by the International Continence Society (ICS) is any complaint of involuntary loss of urine, with an isolated report of loss

Correspondence: Caroline Wanderley Souto Ferreira, Centro de Ciências da Saúde, Departamento de Fisioterapia, Universidade Federal de Pernambuco, Avenida Professor Moraes Rego, 1235, Cidade Universitária, Recife – PE – CEP: 50670-901, Brazil. E-mail: caroline.wanderley@hotmail.com, caroline.wanderley@ufpe.br

insufficient to diagnose this dysfunction (Abrams et al., 2010). The main forms of UI are the following: stress urinary incontinence (SUI), urgency urinary incontinence (UII) and mixed (MUI) (Abrams et al., 2010). UI can impair quality of life (QOL) and sports performance (Jácome, Oliveira, Marques, & Sá-Couto, 2011).

High-impact sports include activities involving jumping and/or rapid movements generally, and include weight-bearing sports with heavy mechanical loading (Groothausen, Siemer, Kemper, Twisk, & Welten, 1997). These may excessively increase intra-abdominal pressure (IAP), cause pelvic floor muscle (PFM) injuries (Eliasson, Larsson, & Mattsson, 2002; Vitton et al., 2011) and the repeated stretching of the pudendal nerve (Dias et al., 2017). These lesions may give rise to structural deficiencies in PFM, such as stress injuries in the fascia, muscles and/or ligaments, in addition to generating an inadequate transmission of IAP to pelvic floor structures, altering the mechanism of urinary continence and predisposing the subject to UI (Eliasson et al., 2002). Thus, athletics, gymnastics, basketball, volleyball, handball, futsal, combat sports and bodybuilding are risk activities for this musculature (Alves et al., 2017; Bø, 2004; Parmigiano et al., 2014). However, incontinent athletes should not be advised to avoid physical activities (Vitton et al., 2011).

In this way, studies that explore the prior knowledge of athletes about the occurrence of UI in sports, their attitude and practice on this subject, and the prevalence of UI in sports, may guide preventive and curative interventions to minimize the prevalence of this pathology in this context, contributing to greater adherence and performance of young athletes. Thus, this research aims to evaluate the prevalence of UI in female athletes practising high-impact sports and its association with knowledge, attitude and practice.

Methods

Observational and knowledge, attitude and practice (KAP) survey type study conducted with female athletes of athletics, basketball, volleyball, judo, handball and futsal, in the sports centres of the city of Recife – PE, which stood out for the number of athletes and number of annual competitions in the period of March to June 2016.

KAP surveys belong to a category of evaluative studies, called formative assessment. Its purpose is to obtain data on a specific part of the population, identify possible alternatives for an efficient intervention and determine strategies of preventive interventions (MINISTÉRIO DA EDUCAÇÃO, 2002).

The inclusion criteria were: women aged between 18–30 years; practitioners of high-impact sports; with a minimum training time of 1 year, minimum frequency of twice weekly and 1 hour daily (Patrizzi, Viana, Silva, & Pegorari, 2014); participation in frequent competitions; and nulliparous. The excluded criteria were: Athletes who were not active at the time of data collection; presented any disease associated with the appearance of UI; had a history of previous urogynaecological/abdominal surgery; were smokers; used the Pilates method as a complement to physical training; had already performed physiotherapy for UI; lacked fluency in the local language.

The sample calculation used the prevalence of 51.9% of the Thyssen, Clevin, Olesen, and Lose (2002) and the quantitative of 170 high-impact sports athletes, with a sample calculated of 118 athletes, being probabilistic, with a confidence level of 95% and maximum permissible error of 0.05.

This study complied with the standards required by Resolution 466/12 of the National Health Council and was approved by the Human Research Ethics Committee of UFPE, under CAAE 49802115.6.0000.5208. All athletes were informed about the procedures to be performed and signed an informed consent form.

A semi-structured evaluation form was prepared by the researchers to obtain the UI diagnosis and information about sociodemographic data, gynaecological history, urinary loss and type and intensity of the sports practice.

The International Consultation on Incontinence Questionnaire – Short Form (ICIQ-SF), to classify urinary loss for the last 4 weeks and to assess the impact of UI on QOL (Tamanini, Dambros, D’Ancona, Palma, & Rodrigues, 2004), being classified as: none (0 points), mild (1–3 points), moderate (4–6 points), severe (7–9 points) and very severe (10 points) (Riccetto et al., 2005).

The KAP survey, also constructed by the researchers, contained questions related to the athletes’ KAP about the occurrence of UI in high-impact sports. The concepts of KAP were established from previous studies that used this survey (Machado, Sâmico, & Braga, 2012; Marinho, Costa-Gurgel, Cecatti, & Osis, 2003).

- Knowledge: Ability to obtain and possess information to be used, constituting the ability to recall specific facts and apply them in solving problems;
- Attitude: Analyse situations according to predispositions and establish opinions directed to a situation;
- Practice: Application of knowledge and decision-making to carry out the action.

For construction, this survey was submitted to the evaluation of seven specialists in the area of

gynaecology and women's health, using the Delphi method, to verify the relevance of the suggested questions and the concept that was intended to be verified (Perroca, 2011).

Each question of the questionnaire was judged by the specialists through a standardized score (Machado et al., 2012) and these answers were considered for the formulation of the following, with each new phase built on the return of the previous phase, in a continuous process until the verification of at least 70% agreement among the specialists for each issue (Hasson, Keeney, & McKenna, 2000).

The knowledge session presented 10 questions with 3 answers: "true", "false" and "do not know". Knowledge was considered appropriate when the answer was "true" for the correct assertions or "false" for the incorrect assertions and inappropriate when the answer was "false" or "I do not know" for true assertions or "true" or "do not know" for false assertions (Machado et al., 2012).

The attitude session consisted of 10 questions with 5 choices of answers: "I agree", "strongly agree", "disagree", "strongly disagree", and "have no opinion". The attitude was deemed appropriate when the answer was "disagree" or "strongly disagree" with false assertions and "agree" or "strongly agree" with true assertions. And inadequate when the answer was "disagree", "strongly disagree" or "I have no opinion" for the true assertions or "agree", "strongly agree" or "have no opinion" for the false ones (Machado et al., 2012).

The practice session presented eight questions for incontinent athletes and two for continent ones with two dichotomous choices of answers: "yes" and "no" (Machado et al., 2012). The practice was considered appropriate when the answer was "yes" or "no" for the correct answers, depending on the context used in the particular question.

For each correct response regarding knowledge and attitude, a score of 1 was recorded and for the incorrect ones, the score was 0. In practice, for the incontinent athletes, the score was 1.25 for the correct responses and 0 for the incorrect ones. For the continent ones, the score was marked as 5 for the correct responses and 0 for the incorrect ones.

The means of adequate responses for each session were calculated, defining as a cut-off a minimum of 70% accuracy. Thus, the knowledge and attitude sessions with a score ≥ 7.0 were considered adequate and < 7.0 , inadequate. For practice, a correct rate ≥ 6.0 for incontinent athletes and ≥ 2.0 for continent ones was considered adequate, < 6.0 and < 2.0 , respectively, inadequate.

The statistical analysis was performed through the Statistical Package for Social Science (SPSS), Chicago, IL, USA, version 19.0. The descriptive

analysis was performed through the mean and standard deviation for the quantitative variables, and absolute and relative frequencies for the qualitative variables.

The association of UI and its three types were determined with variables related to sport through the Pearson Chi-square test; and the association of UI with the KAP, sociodemographic, gynaecological and sports-related variables through univariate and multivariate analyses. The quantitative variables were transformed into qualitative variables through division by the median. A stepwise strategy was used, leaving the predictor variable with the highest value of p , until only the significant associations ($p < .05$) were left in the final model. The odds ratio (OR) was calculated considering a 95% confidence interval and a significance level of $p < .05$.

Results

From a total of 170 female athletes, 151 were contacted and 8 refused participation (Figure 1). Thus, 143 athletes were evaluated according to the eligibility criteria and 118 were selected for the research. The age of the athletes was 21.6 ± 2.7 (mean $\pm$ standard deviation) years old.

Regarding the presence of UI, 82 athletes (70%) lost urine in some daily situation, with 19 athletes (23%) presenting SUI; 19 (23%), UII; and 44 (54%), MUI (Table I), with a higher frequency of once per month loss ($n = 32$, 50.8%) and of a small amount ($n = 61$, 96.8%).

The distribution of UI prevalence during training and competition, demand for medical and physiotherapeutic care for urinary loss and use of emergency strategies are shown in Table I, presenting a

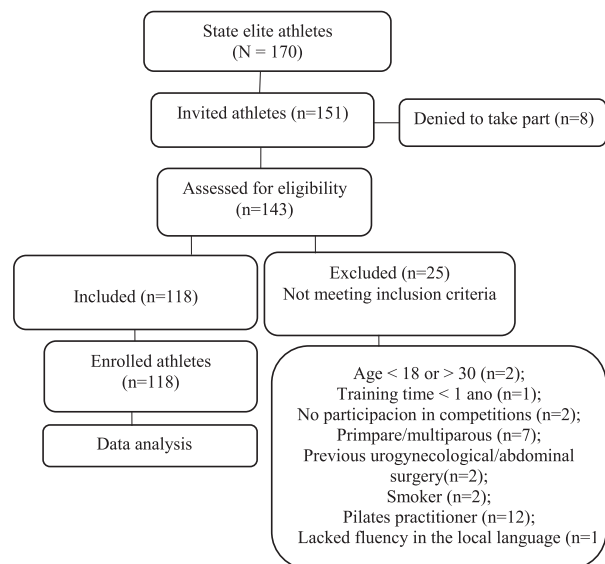


Figure 1. Enrolment flow diagram.

Table I. Distribution of the prevalence of urinary incontinence with variables related to sports.

Variables	UI <i>n</i> (%)	SUI <i>n</i> (%)	UII <i>n</i> (%)	MUI <i>n</i> (%)	<i>p</i> -Value
UI during practice	82 (70)	19(23)	19(23)	44(54)	
UI during competitions	50 (61)	12 (24)	4 (8.)	34 (68)	.000*
Use of emergency strategy	37 (45)	7 (19)	3 (8)	27 (73)	.003*
Use of pad	10 (12)	2 (20)	1 (10)	7 (70)	.48
Hydric restriction	12 (15)	5 (42)	1 (8)	6 (50)	.18
Medical assistance	3 (4)	0 (0)	0 (0)	3 (100)	.15**
Physiotherapeutic assistance	0 (0)	0 (0)	0 (0)	0 (0)	–

Legend: UI – Urinary incontinence; SUI – Stress urinary incontinence; UII – Urgency urinary incontinence; MUI – Mixed urinary incontinence. *Pearson's Chi-square test. **Pearson's Chi-square test for likelihood ratio.

significant difference among the types of UI for the first two variables.

In terms of the prevalence of UI by sports modality, judo was the sport with the highest prevalence of urinary loss ($n = 2$, 100%), followed by athletics ($n = 7$, 87.5%), handball ($n = 22$, 73.3%), basketball ($n = 14$; 67%) and futsal ($n = 20$, 67.0%), and volleyball ($n = 17$; 63.0%), with no significant difference among sports types ($p = .693$).

For ICIQ-SF, 58 athletes (70.7%) lost urine in the last 4 weeks, with a mean of 1.98 points for impact on QOL, reflecting a slight impact.

Regarding the KAP survey, 37 athletes (31%) presented adequate knowledge, 63 (53%) adequate attitude and zero (0.0%) adequate practice. The distribution of adequate responses is shown in Table II.

The association among the prevalence of UI and KAP, socio-demographic and sport-related variables

Table II. Questions answered by the athletes about the KAP of occurrence of urinary incontinence in high-impact sports.

KNOWLEDGE

1. The PF is a structure at the base of the pelvis and is formed by muscles, fascias and ligaments
2. UI is defined as any involuntary loss of urine
3. UI affects only the elderly and is considered a physiological alteration of ageing
4. All urinary loss occurs due to PF muscle weakness
5. Strong PF muscles can prevent UI
6. There are specific strengthening exercises for this musculature
7. High-impact sports such as basketball, volleyball, athletics, handball, among others, have a higher risk of developing UI
8. The activities that provide the greatest loss of urine in high-impact sports are those that include jumping, landing and running
9. UI is not associated with time, duration and frequency of training
10. High-impact sports can cause UI as a result of increased force overload transmitted to PF muscles

ATTITUDE

1. Loss of urine during training does not disrupt performance
2. Loss of urine should be restricted to the athlete. Trainers should not encourage prevention or question athletes about UI
3. Losing urine during the practice of this type of sport is normal, it is not necessary to seek medical or physiotherapeutic care
4. The practice of strengthening PF muscles can prevent the risk of developing UI
5. It is necessary to seek guidance from health professionals about strengthening PF muscles only if there is loss of urine
6. PF muscle strengthening exercises can only be performed in the presence of a physical educator or physiotherapist
7. Strengthening the PF requires a lot of time and money
8. It is better to abandon the sport than to experience the embarrassment of divulging the urinary problem to the trainer
9. To know previously that high-impact sport can present a risk of developing UI could influence the choice of sports practice
10. The risk of developing UI and the embarrassment of urine loss does not compensate for sports practice

PRACTICE

1. Do you lose urine during training?
2. Do you lose urine during competitions?
3. Do you perform strengthening of the PF muscles?
4. Do you discuss about preventing UI in high-impact sports with the trainer or teammates?
5. Do you restrict training due to loss of urine?
6. Do you use absorbents as protection against this urinary loss?
7. Do you decrease the amount of fluid ingested before/during training to avoid losing urine?
8. Do you inform the trainer that you are losing urine during training or competition?
9. Have you ever sought medical care because of a loss of urine?
10. Have you ever sought physiotherapeutic care because of a loss of urine?

Legend: PF = Pelvic floor; UI = Urinary incontinence.

Table III. Association of UI prevalence with variables analysed.

Variables	With UI <i>n</i> (%)	Without UI <i>n</i> (%)	Univariate analysis		Multivariate analysis	
			OR [CI 95%]	<i>p</i> -Value	OR [CI 95%]	<i>p</i> -Value
<i>Knowledge</i>						
Adequate	21 (26)	16 (44)	0.4 [0.19;0.98]	0.045		
Inadequate	61 (74)	20 (26)				
<i>Attitude</i>						
Adequate	43 (52)	20 (56)	0.9 [0.4;1.9]	0.76		
Inadequate	39 (48)	16 (44)				
<i>Practice</i>						
Adequate	3 (4)	36 (100)	0.0 [0.0;.]	—**		
Inadequate	79 (96)	0 (0)**				
<i>Type of sport</i>						
Athletics	7 (8)	1 (3)	1.0 [—]	0.85		
Handball	22 (27)	8 (22)	0.4 [0.04;3.7]	0.42		
Volleyball	17 (21)	10 (28)	0.2 [0.03;2.3]	0.22		
Basketball	14 (17)	7 (19)	0.3 [0.03;2.8]	0.28		
Futsal	20 (24)	10 (28)	0.3 [0.03;2.6]	0.27		
Judo	2 (2)	0 (0)	2.3E8 [0.00;.]	0.999		
<i>Time SP</i>						
≤8 years	49 (60)	14 (39)	1.0 [—]	0.038	2.7 [1.4;4.98]	0.002*
>8 years	33 (40)	22 (61)	0.4 [0.2;0.96]			
<i>Training days/week</i>						
≤3	54 (66)	23 (64)	1.0 [—]	0.84		
>3	28 (34)	13 (36)	0.9 [0.4;2.1]			
<i>Training hours/day</i>						
≤2 hours/day	64 (78)	28 (78)	1.00 [—]	0.97		
>2 hours/day	18 (22)	8 (22)	0.98 [0.4;2.5]			
<i>Age</i>						
≤21 years	50 (61)	16 (44)	1.00 [—]	0.098		
>21 years	32 (39)	20 (56)	0.5 [0.2;1.1]			
<i>Education</i>						
≤17 years of schooling	49 (60)	19 (53)	1.00 [—]	0.48		
>17 years of schooling	33 (40)	17 (47)	0.8 [0.3;1.7]			
<i>BMI</i>						
Low weight	2 (2)	1 (3)	1.0 [—]	0.58		
Normal weight	63 (77)	32 (89)	0.98 [0.1;11.3]	0.99		
Overweight	15 (18)	3 (8)	2.5 [0.2;37.3]	0.51		
Not answered	2 (2)	0 (0)	8.1E8 [0.0;.]	0.999		
<i>Profession</i>						
Student	62 (76)	29 (81)	1.0 [—]	0.86		
Athlete	6 (7)	3 (8.3)	0.9 [0.2;4.0]	0.93		
Teacher	6 (7)	2 (5.6)	1.4 [0.3;7.4]	0.69		
Other	8 (10)	2 (6)	1.9 [0.4;9.4]	0.45		
<i>Marital status</i>						
Single	78 (95)	33 (92)	1.00 [—]	0.47		
Married	4 (5)	3 (8)	0.6 [0.1;2.7]			
<i>Sexually active</i>						
Yes	69 (84)	27 (75)	1.8 [0.7;4.6]	0.244		
No	13 (16)	9 (25)				
<i>Dyspareunia</i>						
Yes	29 (35)	6 (17)	2.7 [1.0;7.3]	0.046	2.99 [1.2;7.6]	0.021*
No	53 (65)	30 (83)				

Legend: UI = Urinary incontinence; *n* = Relative frequency; % = Absolute frequency; OR = odds ratio; CI = confidence interval; binary regression and multiple regression analysis; **p*<0.05; **Not submitted to statistical test for presenting a quad without observation; BMI = Body mass index; Time SP = Time of sports practice.

is shown in Table III, with significant values found only for the variables of adequate knowledge, sports practice time >8 years and dyspareunia in the univariate analysis, and the two latter ones, in the multivariate one.

Discussion

The results of this study demonstrated a high prevalence of UI in female athletes, with MUI being more prevalent in daily situations, during training

and competitions, differing from previous studies (Almeida et al., 2016; Alves et al., 2017; Carls, 2007; Eliasson, Nordlander, Mattsson, Larson, & Hammarstrom, 2004; Jácome et al., 2011; Simeone et al., 2010).

According to Almeida et al. (2016), athletes have three times more chance of suffering from UI than non-athletes. This high prevalence can be explained by the constant increase in the IAP that these athletes undergo during sports practice, which can generate structural injuries in the ligaments, fascia and muscles and thus alter the mechanism of urinary continence (Bø, 2004; Eliasson et al., 2004; Vitton et al., 2011). In addition, athletes from high-impact sports are more likely to have UI than athletes from low-impact sports (Alves et al., 2017; Nygaard & Shaw, 2016); and because of this, athletes from high-impact sports were selected to take part on this research.

PFM fatigue can also lead to changes in the athletes' pelvic floor. Women with SUI have worse performance for fast contractions over a long period when compared to continent women, verified in the resistance test with fast contractions. This contraction aims to improve the urethral response during abrupt exertion activity and to prevent SUI. However, this difference was not verified for sustained contractions. However, women with SUI report higher urinary loss at night, suggesting that PFM cannot maintain muscle contraction capacity for prolonged periods (Burti, Hacad, Zambon, Polessi, & Almeida, 2015).

Both justifications can provide pelvic floor damage, weaken the PFM, and alter the positioning of these muscles and pelvic organs, such as the bladder and ureterovesical junction, causing sphincter insufficiency and/or hypermobility of the bladder neck and triggering SUI (Bø, 2004), which has already been studied and documented in athletes (Bø, 2004; Eliasson et al., 2004; Parmigiano et al., 2014; Thyssen et al., 2002). It seems that the cases of UUI and MUI in athletes are still poorly investigated and disseminated in the sports and scientific environment.

The high prevalence of MUI in this study can be explained by the mechanisms explained previously, because the lower bladder position may also increase stress on stretch receptors in the lower portion of this organ, generating repetitive feedback in the brain, inducing symptoms of an overactive bladder, and triggering urgency symptoms (Petros & Woodman, 2007).

The highest prevalence of UI was found in individual sports: judo and athletics; however, no significant difference was observed among sports. This result was probably due to small numbers of athletes in the judo and athletics group; in view of the fact that long jumps, frequent in athletics, present the highest occurrence of UI because they are the ones

that generate the most pressure on the pelvic floor (Bø, 2004), favouring urinary loss in this sport.

The prevalence of UI by type of sport, practice in hours per day and days per week was not significant, as well as in a similar study (Jácome et al., 2011), but these results differ from Alves et al. (2017) that found a positive association between the volume of exercise, measured in minutes per week, and the frequency of urinary loss. However, the time of sports practice in years presented a significant relationship with UI in this study, corroborating with a similar study (Eliasson et al., 2002), and presenting a little over 2.5 times more chance to develop UI in sports practice greater than 8 years, confirming that intense and prolonged effort affects the mechanism of urinary continence (Simeone et al., 2010).

Dyspareunia was verified in most of the athletes, presenting a significant relation with UI. A recent study (Coksuer et al., 2011) analysed the effect of UI on sexual function in women aged 35–50 years with symptoms of the three types of UI and found that MUI was the one with the greatest impact on sexual function (Coksuer et al., 2011). In addition, pathological viscera-muscle reflexes may culminate in muscle instability and hypertonic contraction, resulting in decreased PFM function, which may generate UI, as well as the development of myofascial syndrome and generation of tension points (Mense, 1993), and may explain the presence of dyspareunia and genitourinary complaints in these athletes.

Despite the high prevalence of UI, few athletes sought medical attention, none sought physiotherapeutic care and none reported urinary loss to the trainer. However, the vast majority, in our study, believe that UI should not be restricted to athletes and that the trainer should encourage prevention. Athletes do not usually talk about urinary loss (Borba, Lelis, & Brêtas, 2008; Carls, 2007; Parmigiano et al., 2014) or seek medical attention (Jácome et al., 2011; Thyssen et al., 2002), probably because of embarrassment (Borba et al., 2008) and insufficient information about treatment and prevention options (Carls, 2007; Parmigiano et al., 2014).

Most athletes judged that urinary loss did not interfere with athletic performance, and incontinent athletes were not concerned about this loss, unlike results from other studies (Carls, 2007; Jácome et al., 2011). However, these athletes also did not seek medical attention to reverse the problem (Almeida et al., 2016; Carls, 2007; Jácome et al., 2011), using emergency strategies to circumvent this situation (Almeida et al., 2016; Jácome et al., 2011; Thyssen et al., 2002). In our study, these strategies were poorly used, with a higher percentage for the use of water restriction, as well as in a similar study (Jácome et al., 2011), and probably due to

the thought that reducing the amount of urine would also decrease the chance of losing urine (Jácome et al., 2011).

Athletes reported only a slight impact on QOL from UI. Despite the concern with UI, athletes believe that this is a normal condition, is not related to sports practice, and does not negatively affect QOL (Jácome et al., 2011). The majority of the urinary loss occurred in small amounts and frequency in the present study, as well as in other studies (Bø, 2004; Bø & Sundgot-Borgen, 2001; Carls, 2007; Eliasson et al., 2004; Jácome et al., 2011) making it possible to justify why athletes do not consider UI to affect QOL (Jácome et al., 2011). However, urinary loss is an individual condition and may differ among women, making possible a negative effect on social life, sports, and exercise (Carls, 2007).

A low level of adequate knowledge and practice was detected in the athletes of this study, although most correctly respond that stronger PFM can prevent UI and that there are specific strengthening exercises for this musculature. In contrast, few athletes are aware of the relationship between high-impact sports and UI and few perform these strengthening exercises, resulting in a large percentage losing urine in training and competitions. The athletes do not present knowledge of the relationship between sports and UI and its prevention and treatment (Carls, 2007; Jácome et al., 2011), and the increased risk of developing UI even without the most common risk factors (Parmigiano et al., 2014).

The practice of strengthening PFM is poorly performed by athletes (Carls, 2007; Thyssen et al., 2002), as in our study, due to the lack of knowledge of the existence of these exercises, but with the affirmation of their interest in performing them if they were instructed to perform them (Carls, 2007). The high prevalence of UI implies that the strengthening of PFM should be routinely included in training programmes (Bø, 2004), since these exercises are recommended as the first treatment for MUI and SUI (Wilson et al., 2005).

Adequate knowledge presented a 57% lower chance of developing UI. It means that, the greater dissemination of information about the relationship between UI and sports, its prevention and treatment, will reduce the prevalence of this dysfunction in sports.

As for the practice, it was verified that athletes lose more urine during training than competitions, as verified in another study (Thyssen et al., 2002), and can be justified by the use of water restriction and bowel emptying before competitions or altered diuresis (Thyssen et al., 2002). In addition, during competitions there is a higher urinary catecholamine level when compared to training (Baron et al., 1992), thus, because it has α -receptors, the urethra

remains closed, preventing the loss of urine during competitions (Thyssen et al., 2002).

No athlete presented adequate practice, which is a plausible reason for the high prevalence of UI found. However, due to the absence of athletes with adequate practice (Table III), it was not possible to run any statistical test to confirm this relationship.

It is recommended for future training practice, to introduce preventive physiotherapy with the aim of evaluating the PFM of all athletes to check if they already have any impairment of the PFM and/or pelvic organs; showing the importance of the PFM for athlete's health and performance; as well as, teaching exercises to strengthen fast and slow twitch muscle fibres of the PFM.

It is recommended for future research, to evaluate the anatomical position of the pelvic organs; and measurements for pelvic floor dysfunction in incontinent athletes through translabial ultrasound, to prove that female high-impact athletes, without proper PFM training, have greater bladder neck descent and lower bladder position that are related to UI; as well as, assess the strength and endurance of the PFM.

Conclusion

A prevalence of 70% of UI was observed, with a predominance of MUI (54%), and a low level of knowledge and adequate practices of athletes on the occurrence of UI in sports. Sports practice over 8 years and dyspareunia were predisposing factors to UI by approximately 2.5 and 3 times, respectively. Adequately trained athletes had a 57% lower chance of developing UI, and incontinence had a slight QOL impact in affected athletes.

Thus, greater attention is needed on the urogynecologic aspects of female high-impact athletes; as well as, a greater dissemination of knowledge and preventive practices about UI in sports for decreasing the prevalence of UI in this environment and generating a greater adherence of young athletes to sports practice.

Disclosure statement

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ORCID

Caroline Wanderley Souto Ferreira  <http://orcid.org/0000-0003-2736-5918>

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